KYAMBOGO UNIVERSITY

Final Year Project:

Department of Electrical and Electronics Engineering

GSM Signal Denoising and Modulation Classification for Rural Uganda

Improving Connectivity in Underserved Areas Using Machine Learning

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Proposal Outline

Background

- UCC reports that network subscriptions reached **33.2 million** by the end of 2022, and mobile data traffic doubled from 217 million GB in 2020 to 421.5 million GB in 2022 showing an exponential growth in network service demand. [1]

- Unfortunately, network expansion is failing to keep pace with this demand; only 31% of the population and 24% of the landmass have 4G coverage and 62% of the population (30 million people) still lacks mobile internet access, and live in low SNR, high-interference environments where the signal is weak (below -90 dBm), the noise becomes louder than the message, making it impossible for standard tools to understand what is being sent[3][4]

- Many people in rural Uganda still use basic phones for calls and Mobile Money, but the signal is often very weak and are being blocked or "noised up" by illegal boosters, unlicensed community radios (Bizindaalo), and heavy tropical rain.

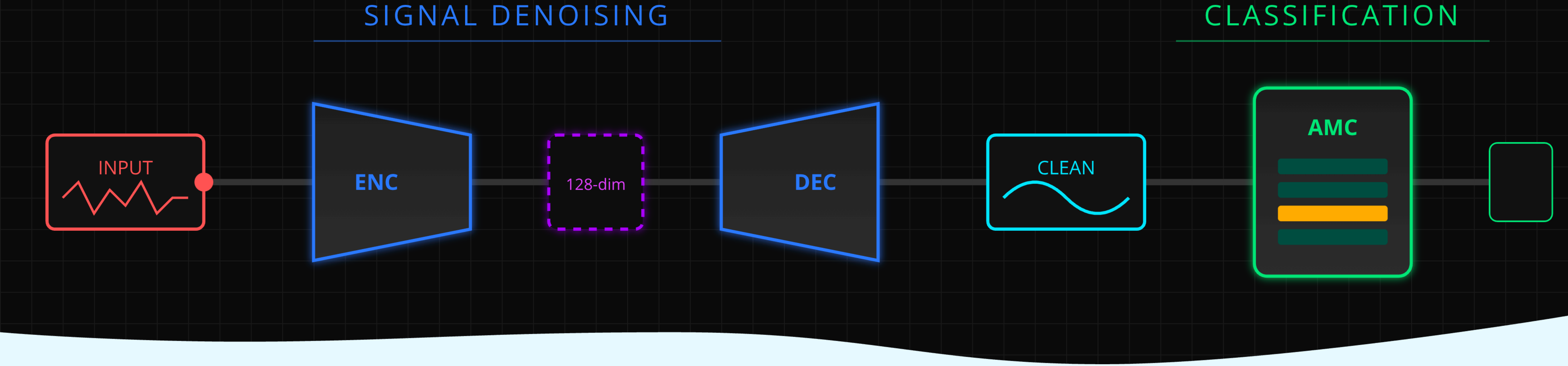
Background continuation

- Automatic Modulation Classification (AMC) is crucial for regulators to monitor the spectrum, manage interference, and identify illegal broadcasters that threaten critical services like aviation.
- We propose using a Denoising Autoencoder (DAE) combined with an AMC system that takes a marginal signal (e.g., at -90 dBm or even lower) and actively cleans it up removing noise before trying to classify or demodulate it.
- Instead of just trying to read a noisy signal, our system first cleans the signal to make it clear again making sure that even in the most remote sub-counties, the network can still identify and process signals correctly.

Problem Statement

- Many people in rural Uganda still use basic phones for calls and Mobile Money, but the signal is often very weak. When the signal is weak (below -90 dBm), the noise becomes louder than the message, making it impossible for standard tools to understand what is being sent

- Other signals are being blocked or noised up by illegal boosters, unlicensed community radios (Bizindaalo), and heavy tropical rain. The Uganda Communications Commission (UCC) finds it hard to catch illegal broadcasters because the current tools cannot see through all this noise. This leads to dropped calls, failed Mobile Money transactions, and poor service for over 30 million Ugandans. Over 70% of call failures in some areas are due to this interference, yet we lack a smart way to clean the signal before processing it.



Main Objective

Design a Hybrid DAE–AMC Pipeline

To design, implement, and empirically evaluate a hybrid **Denoising Autoencoder–Automatic Modulation Classification (DAE–AMC) pipeline** that suppresses interference in noisy I/Q streams and maintains accurate modulation decisions across low SNR channel conditions observed in Ugandan networks.

Specific Objectives

01

Review AMC and DAE literature emphasizing low SNR mitigation strategies relevant to Ugandan channel conditions.

02

Architect and train Conv1D DAE module reconstructing clean I/Q samples from noisy inputs at –90 dBm thresholds. Implement supervised AMC classifier ingesting raw and DAE-cleaned signals, quantifying feature preservation.

03

Benchmark integrated DAE–AMC against standalone AMC using accuracy, F1-score, confusion matrices across various SNR.

Justification and Significance

- **Meeting UCC Mandates:** The DAE is explicitly trained on SNR conditions matching UCUSAF's –90 dBm coverage floor, addressing the "near-far" interference documented by UCC QoS audits rather than generic low-SNR scenario

- **It Aligns with National Policy:** Our project supports Uganda's national development plans (NDPIII/NBP) that call for more spectrum-efficient technologies. It replaces older, "brittle" handcrafted methods with a robust, cost effective, software-based AI-based algorithm for enhancing signals by removing noise.

Spectrum Surveillance

It will help with the detection of illegal transmitters and interference mitigation for aviation and public-safety channels.

Rural Broadband

It will help maintains reliable service in low-SNR environments, supporting 30M Ugandans lacking mobile internet access.

Critical Services

Ensures stability for mobile money and e-government services dependent on reliable radio links.

Scope

Offline Experimentation

Project conducted entirely on a computer using pre-existing data, ensuring a controlled environment.

Public Datasets

DeepSig RadioML 2018.01A benchmark (synthetic, well-characterized SNR labels) and the RF Signal Data collection on Kaggle (real SDR captures providing hardware impairment realism) [11], [12]

Noise Modelling

We inject Uganda-specific noise profiles such as Booster oscillation noise as High-power wideband impulses simulating illegal repeater feedback, Bizindaalo harmonics as Narrowband tones at 2nd/3rd harmonics of FM frequencies (200 MHz, 300 MHz sidebands leaking into 900 MHz), Rain fade attenuation as Time-varying SNR reduction modeling tropical storm conditions (100+ mm/h rainfall rates)

Model Development

Focus on a Conv1D Denoising Autoencoder (DAE) and a supervised AMC classifier.

Specific Modulations

Training models to recognize GSM-family modulations like GMSK, GFSK, and QPSK

Platform & Tools

Windows OS, Python, TensorFlow, Keras, Jupyter Notebook, and Google Cloud Platform.

Timeline

A 6-month dedicated period for research, model development, and comprehensive testing.

Literature Review

Chapter Two

Classical Methods

Early Automatic Modulation Classification (AMC) relied on statistical analysis and manually engineered features. While elegant, these methods are brittle and fail completely in the low-signal, high-interference environments common in Uganda.

Key Limitation:

- They are not robust under unknown channels
- These impairments are documented by UCC in congested Ugandan deployments motivates the transition toward data-driven feature learning.

Machine learning is the new standard

Modern research has shifted to using deep learning (DL) models like CNNs, Transformers, and Mixture-of-Experts frameworks. These models can learn features directly from raw signal data and have shown significant accuracy improvements, especially in low Signal-to-Noise Ratio (SNR) conditions down to 0 dB.

GSM Noise Mitigation in African Networks

While DAE-based signal enhancement has been extensively studied academically, its application to African GSM networks' specific interference environment remains unexplored.

Regulatory Interventions

UCC's zero-tolerance posture: enforcement operations confiscate illegal boosters and shut down unlicensed Bizindaalo stations. Proliferation continues due to porous borders and high demand.

Technical Countermeasures

Modern 4G/5G use Massive MIMO and beamforming, but unavailable for legacy 2G/GSM in rural areas. NBI fiber expansion reduces microwave backhaul reliance.

The Research Gap

No published work applies ML-based denoising to UCC-documented GSM interference: booster oscillation, Bizindaalo harmonics, and tropical rain fade.

> **We fill that gap** by training a DAE on synthetic noise profiles matching documented interference vectors, enabling GMSK classification at SNR levels where conventional approaches fail.

Machine Learning in AMC

Deep learning has reshaped AMC by learning discriminative features directly from raw I/Q samples. Recent contributions demonstrate tangible low-SNR gains:

1

AbdElaziz et al. (2023)

Robust CNN with parallel asymmetric kernels achieved 96.5% accuracy at 0 dB and 86.1% at −2 dB across nine modulations.

2

Zhang et al. (2023)

MoE-AMC mixture-of-experts framework yielded ~71.8% averaged accuracy across −20…18 dB—about 10% higher than single-expert models.

3

Meta-Transformer (2024)

Leverages transformer encoders and few-shot learning to adapt rapidly to unseen modulations while maintaining superior accuracy.

4

Rehman et al. (2025)

DLAMC converts I/Q streams into eye diagrams, overcoming the 10–48% accuracy ceiling near −10 dB SNR.

5

Jagannath et al. (2022)

Validated CNN-based multitask AMC on USRP SDR testbed, demonstrating >98% accuracy in live over-the-air experiments.

Denoising Autoencoders for Signal Enhancement

Denoising frontends have emerged as an effective countermeasure when raw I/Q features are overwhelmed by interference.

75%

Zhang et al.

Dual residual DAE with channel attention improved AMC accuracy by up to 75% across −12…8 dB SNR.

77.5%

DenoMAE

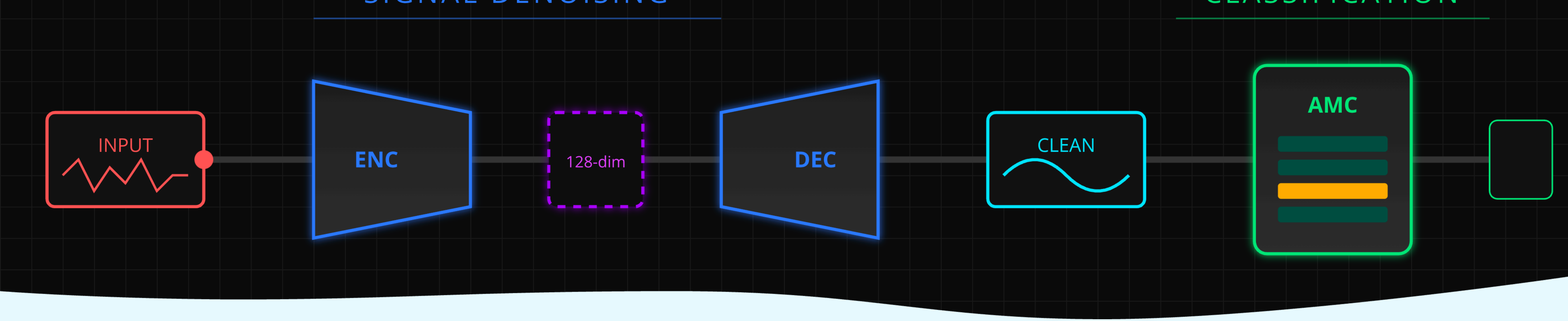
Faysal et al.'s multimodal denoising masked autoencoder sustained 77.5% accuracy at −10 dB—roughly 22% higher without denoising.

70%

Thresholded AE

An and Lee's thresholded autoencoder with SNR predictor delivered ~70% relative accuracy gains on low-SNR samples.

These findings justify the DAE–AMC architecture explored in this project and provide design cues for gating strategies that conserve energy on SDR deployments.



CHAPTER THREE – METHODOLOGY

A comprehensive framework for developing robust Automatic Modulation Classification systems tailored to challenging RF environments through deep learning and denoising techniques.

Introduction to Methods

We adopt an experimental research design anchored in reproducible data processing and quantitative benchmarking across four interconnected pillars:

01

Dataset Curation

Aggregate I/Q datasets from simulated and real captures from RF Signal Data and RadioML 2018.01A

02

Controlled SNR Scenarios

Generate SNR conditions (−12 to +18 dB) mimicking edge conditions with added impairments: Rician fading, frequency offsets, and timing jitter.

03

Model Training

Train Conv1D denoising autoencoder (unsupervised) and 1D CNN AMC classifier (supervised) both independently and as integrated pipeline.

04

Comparative Evaluation

Compare hybrid system against standalone baselines using accuracy curves, F1-scores, and confusion matrices for SDR deployment readiness.

Research Strategy

Dataset Harmonization

Convert datasets to unified tensor format (1024-sample length, normalized amplitude) with stratified train/validation/test splits by modulation and SNR.

Uganda Specific Noise Modeling

Inject three UCC-documented interference types: booster oscillation (wideband impulse), Bizindaalo harmonics (narrowband FM tones), and rain fade attenuation.

DAE Pretraining

Train Conv1D encoder–decoder on noisy/clean pairs using MSE loss until reconstruction PSNR converges, ensuring low-SNR fidelity.

AMC Training

Train baseline on raw GSM-family modulations (GMSK, GFSK, QPSK) inputs, then retrain with DAE outputs to create hybrid pipeline, comparing frozen and fine-tuned configurations.

Cross-Dataset Evaluation

Assess models on held-out SNR bins, unseen modulation families, and OTA-style splits to expose and quantify domain shift effects.

Deployment Prototyping

Integrate trained models into lightweight inference service with GUI for visualization and regulatory demonstrations.

Project Budget Overview

The successful execution of this project requires a structured allocation of resources. The total estimated budget is 800,000 UGX, covering critical areas from software to research.

Item	Cost (UGX)
Software Licenses	200,000
Data Curation	100,000
Virtual Machines	200,000
Cloud Databases	100,000
Internet and Research	50,000
Printing and Documentation	50,000
Miscellaneous	100,000
Total Estimated Cost	**800,000**

This budget ensures that all necessary technical infrastructure, data processing capabilities, and operational requirements are adequately funded to support the project's objectives.

Project Timeline

Our project will be executed in five distinct phases, ensuring a structured and efficient progression towards our objectives over a 14-week period.

Phase	Description	Weeks	Timeline (Week)
Phase 1: Initiation & Literature Review	Define problem, scope, objectives, and conduct literature review.	1-2	
Phase 2: Data Acquisition & Preprocessing	Gather datasets, implement preprocessing pipelines, and perform feature extraction.	3-5	
Phase 3: Model Development & Training	Select & implement ML algorithms, develop models, train & validate.	6-9	
Phase 4: Testing & Optimization	Evaluate model performance, fine-tune hyperparameters, and optimize.	10-12	
Phase 5: Documentation & Presentation	Compile research findings, methodologies, results, and prepare presentation.	13-14	

This timeline ensures a methodical approach to each stage, culminating in a robust and well-documented solution for intelligent signal processing.

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Thanx for Listening